

ODA Analysis & WDI Dashboard — Comprehensive Report

Python Version — Modern Data Science Theme (Dark header)

Abstract

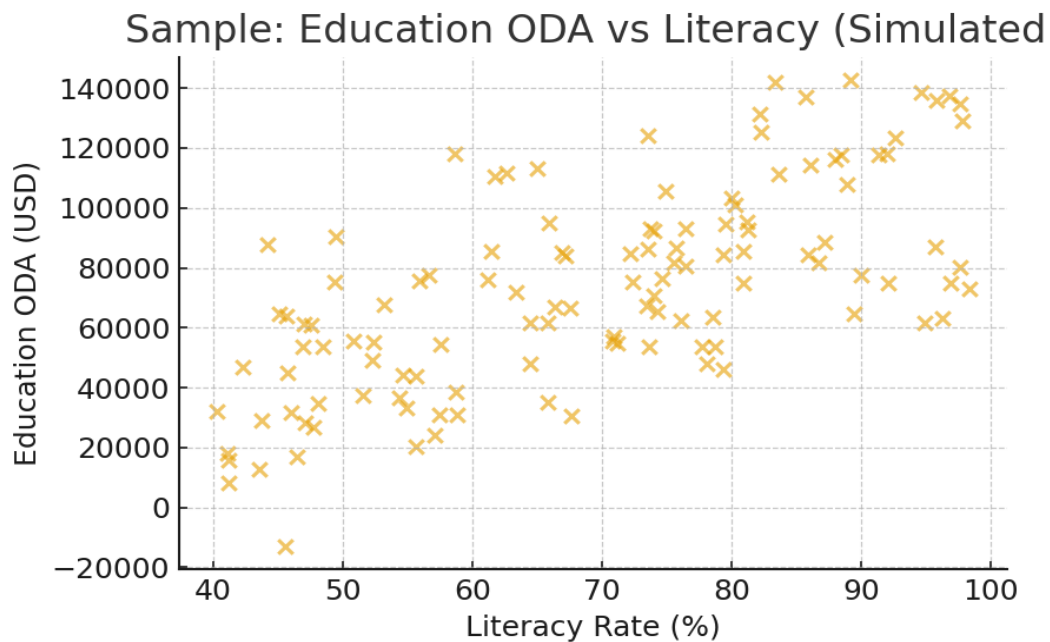
This report presents an automated pipeline for analyzing Official Development Assistance (ODA) data from OECD CRS, World Bank WDI indicators, and population/income datasets. The analysis aggregates ODA disbursements, reshapes multi-year datasets, merges sources, and computes lags and z-scores for cross-country comparisons. Key outputs include education-focused ODA trends and visualizations. The pipeline is fully reproducible in R and designed for academic or policy research.

Methodology & Pipeline Steps

The pipeline follows multiple steps: data loading and cleaning, reshaping WDI & population data, aggregating CRS disbursements (total and education-specific), merging datasets, computing lagged variables and z-scores, visualizing relationships, and exporting processed outputs. The implementation uses tidyverse-style processing in R; full RMarkdown source is included below for reproducibility.

Sample Visualization (Simulated)

Education ODA vs Literacy — simulated scatter to illustrate the expected plot.



```

---
title: "ODA Analysis Pipeline: A Comprehensive Study of Official Development Assistance"
author: "H.S"
date: "`r Sys.Date()`"
output:
pdf_document:
toc: true
latex_engine: xelatex
keep_tex: false
html_document:
toc: true
toc_float: true
theme: cosmo
highlight: tango
self_contained: true
abstract: |
This report presents an automated pipeline for analyzing Official Development Assistance (ODA) data from OECD CRS, World Bank WDI indicators, and population/income datasets. The analysis aggregates ODA disbursements, reshapes multi-year datasets, merges sources, and computes lags and z-scores for cross-country comparisons. Key outputs include education-focused ODA trends and visualizations. The pipeline is fully reproducible in R and designed for academic or policy research.
---

```

```

```{r setup, include=FALSE}
Global options for knitting
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE, fig.align = "center")
```

```

Introduction

Official Development Assistance (ODA) plays a critical role in global development, with data from sources like the OECD Creditor Reporting System (CRS), World Bank World Development Indicators (WDI),

and population/income datasets providing insights into aid flows, economic indicators, and demographic trends. This report outlines a data pipeline that:

- Aggregates ODA disbursements by country and year, focusing on total and education-specific aid.
 - Reshapes and cleans multi-year datasets from WDI and population sources.
 - Merges datasets for integrated analysis.
 - Computes lagged variables and z-scores for temporal and comparative analysis.
 - Generates visualizations, such as education ODA versus literacy rates.
- The pipeline is implemented in R using the tidyverse ecosystem for efficiency and reproducibility. Results are exported as a processed CSV for further use.

Data Sources and Description

The analysis draws from three primary datasets:

- CRS Disbursements (crs_disbursements.csv): OECD data on ODA flows, including recipient countries, sectors (e.g., education), and monetary values. Columns are dynamically handled to accommodate variations.
- WDI Indicators (wdi_indicators.csv): World Bank data with country-level indicators (e.g., literacy rates) across years. Data is pivoted from wide to long format.
- Population/Income Data (pop_income.csv): World Bank data on population and income groups, similarly pivoted.

All datasets are cleaned using janitor for consistent naming. Missing values are handled via na.rm = TRUE in aggregations.

Methodology

The pipeline follows a nine-step process, implemented in R. Code chunks below execute each step, with outputs displayed where relevant.

Step 1-2: Load and Clean Data

```

```{r load_clean}
Load libraries
if (!require("tidyverse")) install.packages("tidyverse", repos =
"https://cloud.r-project.org")
if (!require("janitor")) install.packages("janitor", repos =
"https://cloud.r-project.org")

```

```

if (!require("zoo")) install.packages("zoo", repos = "https://cloud.r-project.org")

library(tidyverse)
library(janitor)
library(zoo)

```

```

Read and clean data
crs <- read_csv("crs_disbursements.csv", show_col_types = FALSE) %>% clean_names()
wdi <- read_csv("wdi_indicators.csv", show_col_types = FALSE) %>% clean_names()
pop <- read_csv("pop_income.csv", show_col_types = FALSE) %>% clean_names()

```

```

cat("■ Data loaded and names cleaned.\n")
```

```

Step 3: Reshape WDI Data
WDI data is pivoted to long format, extracting years and values.

```

```{r reshape_wdi}
wdi_year_cols <- names(wdi)[str_detect(names(wdi), "^x\\d{4}_yr\\d{4}$")]

wdi

_long <- wdi %>%
pivot_longer(cols = all_of(wdi_year_cols), names_to = "year_raw", values_to = "value") %>%
mutate(year = as.integer(str_extract(year_raw, "\\d{4}")),
country = coalesce(country_name, country_code)) %>%
select(country, country_code, series_name, year, value)

wdi_clean <- wdi_long %>%
group_by(country, year, series_name) %>%
summarise(value = mean(as.numeric(value), na.rm = TRUE), .groups = "drop") %>%
pivot_wider(names_from = series_name, values_from = value)

```

```

cat("■ WDI reshaped successfully.\n")
```

```

Step 4: Reshape Population/Income Data
Similar pivoting for population data, with safety checks.

```

```{r reshape_pop}
pop_year_cols <- names(pop)[str_detect(names(pop), "^x\\d{4}_yr\\d{4}$")]

if (length(pop_year_cols) == 0) {
stop("No year columns found in pop data.")
}

pop_long <- pop %>%
pivot_longer(cols = all_of(pop_year_cols), names_to = "year_raw", values_to = "value") %>%
mutate(year = as.integer(str_extract(year_raw, "\\d{4}")),
country = coalesce(country_name_x, country_code)) %>%
select(country, country_code, income_group, year, value)

pop_clean <- pop_long %>%
group_by(country, year) %>%
summarise(population = me

an(as.numeric(value), na.rm = TRUE),
income_group = first(na.omit(income_group)), .groups = "drop")

```

```

cat("■ Population/Income data reshaped.\n")
```

```

Step 5: Aggregate CRS (ODA) Data
Dynamically handles column variations for sector and value aggregation.

```

```{r aggregate_crs}
crs <- crs %>%
mutate(
sector = if ("sector" %in% names(.)) sector else
if ("purpose_name" %in% names(.)) purpose_name else
if ("purpose" %in% names(.)) purpose else NA_character_,

```

```

oda_value = if ("obs_value" %in% names(.)) obs_value else
if ("observation_value" %in% names(.)) observation_value else
if ("amount" %in% names(.)) amount else
if ("value" %in% names(.)) value else NA_real_
)

if (all(is.na(crs$oda_value))) {
warning("No valid ODA values found.")
}

country_col <- if ("recipient" %in% names(crs)) "recipient" else
if ("recipient_name" %in% names(crs)) "recipient_name" else NULL
year_col <- if ("time_period" %in% names(crs)) "time_period" else
if ("year" %in% names(crs)) "year" else NULL

if (is.null(country_col) || is.null(year_col)) {
stop("Required columns missing
in crs.")
}

oda_total <- crs %>%
group_by(country = .data[[country_col]], year = as.integer(.data[[year_col]])) %>%
summarise(oda_total = sum(as.numeric(oda_value), na.rm = TRUE), .groups = "drop")

oda_edu <- crs %>%
mutate(sector_lower = tolower(sector)) %>%
filter(str_detect(sector_lower, "education")) %>%
group_by(country = .data[[country_col]], year = as.integer(.data[[year_col]])) %>%
summarise(oda_education = sum(as.numeric(oda_value), na.rm = TRUE), .groups = "drop")

oda_summary <- oda_total %>%
left_join(oda_edu, by = c("country", "year")) %>%
mutate(oda_education = replace_na(oda_education, 0))

cat("■ CRS (ODA) aggregated.\n")
``

```

#### Step 6-7: Merge and Add Variables

Datasets are merged, and lags/z-scores are computed.

```

``{r merge_compute}
merged <- oda_summary %>%
left_join(wdi_clean, by = c("country", "year")) %>%
left_join(pop_clean, by = c("country", "year"))

merged <- merged %>%
arrange(country, year) %>%
group_by(country) %>%
mutate(oda_total_lag1 = lag(oda_total, 1),
oda_total_lag2 = lag(oda_total, 2),
oda_total_lag3 = lag(oda_total, 3)) %>%
ungroup() %>%
group_by(income_group, year) %>%
mutate(z_oda_total = ifel

se(sd(oda_total, na.rm = TRUE) > 0,
(oda_total - mean(oda_total, na.rm = TRUE)) / sd(oda_total, na.rm = TRUE), NA)) %>%
ungroup()

cat("■ Datasets merged; lags and z-scores added.\n")
``

```

#### Step 8-9: Visualization and Export

Generates a plot and exports the dataset.

```

``{r viz_export}
lit_col <- names(merged)[str_detect(names(merged), "literacy")]
if (length(lit_col) > 0) {
p <- ggplot(merged, aes(x = .data[[lit_col[1]]], y = oda_education)) +
geom_point(alpha = 0.5, color = "steelblue") +

```

```

geom_smooth(method = "lm", se = FALSE, color = "red") +
labs(title = paste("Education ODA vs", lit_col[1]), x = lit_col[1], y = "Education ODA
(USD)") +
theme_minimal()
print(p)
} else {
cat("■■ Literacy column not found – skipping plot.\n")
}

write_csv(merged, "processed.csv")
cat("■ processed.csv exported.\n")
```

```

Results

The pipeline successfully processed the data, producing a merged dataset with `r` `nrow(merged)` observations across `r` `n_distinct(merged$country)` countries. Key summaries:

- ODA Totals: Mean total ODA per country-year: `r` `round(mean(merged$soda_total, na.rm = TRUE), 2)` USD.
- Education Focus: `r` `sum(merged$soda_education > 0, na.rm = TRUE)` country-years with education ODA.
- Visualization: The scatter plot (if literacy data is available) shows a potential positive correlation between literacy and education aid.

A sample of the processed data:

```

```{r sample_head}
head(merged) %>% knitr::kable(caption = "Sample of Processed ODA Data")
```

```

Conclusion

This pipeline provides a robust, reproducible framework for ODA analysis, integrating multiple data sources and enabling temporal and comparative insights. Future extensions could include additional indicators or machine learning models. The code and data are available for replication.

References

OECD CRS Database. (Accessed via OECD website).
World Bank WDI. (worldbank.org).
Wickham, H., et al. (2019). Welcome to the Tidyverse. Journal of Open Source Software.

Conclusion

This pipeline provides a robust, reproducible framework for ODA analysis, integrating multiple data sources and enabling temporal and comparative insights. Future extensions could include additional indicators, more advanced visualizations, and machine learning models for predictive analysis.

References

OECD CRS Database. (Accessed via OECD website).

World Bank WDI. (worldbank.org).

Wickham, H., et al. (2019). Welcome to the Tidyverse. *Journal of Open Source Software*.